

CCNA 200-301 V2.0

14 Troubleshooting Layer 2

Part II — Switching and Network Access

EXAM TOPICS

2.4

LECTURER

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Where this chapter sits

Part I	Foundations	chapters 1–7
Part II	Switching and Network Access	chapters 8–14
Part III	IP Routing	chapters 15–19
Part IV	Network Services and Security	chapters 20–26
Part V	Wireless, Virtualisation and Operations	chapters 27–32
Part VI	Sitting the Exam	chapters 33–34

What you will be able to do

- **Apply** a repeatable order of checks to a layer 2 fault instead of guessing.
- **Select** the right `show` command for a symptom, and read what it returns.
- **Use** `show logging`, ping, extended ping and capture output together to localise a fault.
- **Diagnose** the six faults that account for most switched-network problems.

Before we start

All of Part II. This chapter assembles Chapter 8 to Chapter 13 into a method.

Vocabulary for this chapter

symptom

hypothesis

scope

errdisable

show logging

severity

divide and conquer

bottom-up

baseline

Each is defined where it first matters — not here.

This chapter is the exam's heaviest verb

This is what the blueprint actually asks for

Topic **2.4** — “troubleshoot basic Layer 2/Layer 3 connectivity and device operations using show commands (including show logs), ping, extended ping, trace route, and packet capture output” — is one of eight *troubleshoot* topics. Together with the eight *configure* topics they are 55% of the paper. The method below is worth more marks than any single technology in this book.

Scope first, then layers



Ask “how many?” before you ask “why?”

The single most useful question is how much is broken, because it tells you where to look before you have run a command.

- **One device** → that port, that cable, that host's configuration.
- **One VLAN, everywhere** → the VLAN itself: does it exist, is it shut, is it allowed on the trunks, does its SVI have an address?
- **Everything on one switch** → that switch's uplink, or spanning tree.
- **Everything, everywhere** → a loop, the root bridge, or a routing or gateway failure.
- **Intermittent, under load** → physical layer, duplex, or a saturated link. Almost never a VLAN.

Then work bottom-up

Once you know the scope, walk the layers from the bottom. Each step is one command and rules out a whole class of cause.

- **Is the port up?** `show interfaces status`
- **Is it clean?** `show interfaces` — errors, duplex, speed
- **Is it in the right VLAN?** `show interfaces switchport`
- **Does the switch see the device?** `show mac address-table`
- **Does the VLAN reach the next switch?** `show interfaces trunk`
- **Is spanning tree forwarding?** `show spanning-tree vlan`

Then work bottom-up (continued)

- **Is there a gateway?** `show ip interface brief`

Resist starting at step 7 because the user said “I cannot reach the internet”.

The first three commands



show interfaces status

SW1#

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/1	Uplink to SW2	connected	trunk	a-full	a-1000	10/100/1000
Gi0/5	Desk 5	connected	10	a-full	a-100	10/100/1000
Gi0/6	Desk 6	notconnect	10	auto	auto	10/100/1000
Gi0/7	Desk 7	err-disabled	10	auto	auto	10/100/1000
Gi0/8	Desk 8	connected	15	a-full	a-1000	10/100/1000

show logging | include LINK|LINEPROTO|ERR|SPANTREE

SW1#

```
%LINK-3-UPDOWN: Interface GigabitEthernet0/6, changed state to down
%SPANTREE-2-BLOCK_BPDUGUARD: Received BPDU on port Gi0/7 with BPDU Guard
    enabled. Disabling port.
%PM-4-ERR_DISABLE: bpduguard error detected on Gi0/7, putting Gi0/7 in
    err-disable state
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on
    GigabitEthernet0/1 (999), with SW2 GigabitEthernet0/1 (1).
```

Read the log before you read anything else

The log frequently contains the answer in plain English, timestamped. Above, it names the errdisable cause and a native VLAN mismatch nobody had reported — a fault that is silently bridging VLAN 999 and VLAN 1 between two switches.

Filter it. `show logging | include` with the keywords above turns hundreds of lines into the handful that matter.

The six faults, and what each looks like



Working a fault end to end

Symptom. “Nobody in accounts can reach the file server since this morning. Sales are fine.”

Scope. One VLAN, multiple users, more than one switch. That rules out a single cable and points at the VLAN or its path.

- `show interfaces status` on the access switch — accounts ports are `connected` in VLAN 30. Not a port problem.
- `show mac address-table vlan 30` — addresses are being learned locally. The devices are alive and the switch sees them.

Working a fault end to end (continued)

- `show interfaces trunk` on the uplink — VLAN 30 appears in “allowed” but *not* in “forwarding and not pruned”.
- `show spanning-tree vlan 30` — the uplink is `Altn BLK`.

Working a fault end to end (continued)

- `show logging | include SPANTREE` — a topology change overnight; a second switch was added and won the root election for VLAN 30.

Cause. Not a VLAN fault at all. The new switch became root for VLAN 30 only, which moved the tree and put the accounts uplink into blocking. Sales, in VLAN 20, still had the old root and were unaffected — which is exactly why only one VLAN broke.

Fix. Set the intended root explicitly for all VLANs, add root guard on the ports facing access switches, and confirm the uplink returns to `Desg FWD`.

Common pitfalls

- **Changing something before you have measured.** If you reconfigure before capturing the current state you can no longer tell whether you fixed it or moved it.
- **Changing more than one thing at once.** Then you do not know which change worked, and you cannot write it up.
- **Believing the reporter's diagnosis.** "The switch is broken" usually means "my thing does not work". Take the symptom, discard the theory.
- **Ignoring what still works.** The working half of the network is evidence. Sales working and accounts failing localised the fault above faster than any command did.
- **Skipping the log.** It is timestamped and often explicit.

Two symptoms, one cause, and a misleading first look.

Users report intermittent slowness. One user cannot connect at all. Both started after a contractor visited the comms room.

Two symptoms, one cause, and a misleading first look.

What would you check first — and what do you expect to see?

show interfaces status | exclude connected

SW1#

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/7	Desk 7	err-disabled	10	auto	auto	10/100/1000

show logging | include ERR_DISABLE|MACFLAP

SW1#

```
%PM-4-ERR_DISABLE: bpduguard error detected on Gi0/7, putting Gi0/7 in  
err-disable state  
%SW_MATM-4-MACFLAP_NOTIF: Host 0011.2233.4455 in vlan 10 is flapping  
between port Gi0/7 and port Gi0/12
```

What is the fault — and what does this output rule out?

Why it is doing that

Diagnosis. Both symptoms are one event. The contractor plugged an unmanaged switch into `Gi0/7` and, to “make sure it worked”, also patched it into `Gi0/12`. That created a loop: the MAC flap is the loop being seen from two ports, and the intermittent slowness is the resulting flooding. BPDU guard then errdisabled `Gi0/7`, which is the one port that is definitively dead — and which, by shutting, largely saved the network.

And the lesson

Fix. Remove the second patch lead. Confirm the flapping stops. Then recover the port with `shutdown / no shutdown` — *after* the cause is gone, not before, or it will simply errdisable again. Finally, note that BPDU guard did its job: without it this would have been a full broadcast storm rather than one dead port.

Fix five faults against the clock

Topology. Three switches, one router, six PCs across three VLANs — built, working, and then broken by a partner.

1. Build the network and confirm it works fully. **Capture a baseline:** `show interfaces status`, `show interfaces trunk`, `show spanning-tree vlan` and `show ip interface brief` from every device. You cannot troubleshoot without knowing what right looks like.
2. Hand the network to a partner, who introduces **five** faults from the table above without telling you which.
3. Find and fix them, writing down for each: the symptom, the scope, the command that proved it, and the fix. Give yourself twenty minutes.
4. Swap roles.

Fix five faults against the clock (continued)

5. Compare notes: which fault took longest, and would starting with scope rather than commands have found it faster?
6. **Extension.** Introduce a fault that produces *no* log message — a wrong access VLAN is a good one — and discuss why the log is a first stop rather than a complete answer.

CHECKPOINT 1 of 3

Users on one VLAN across three switches lose connectivity; other VLANs are fine. What are the two most likely causes, and which command separates them?

Discuss · then advance

Checkpoint 1 of 3

Users on one VLAN across three switches lose connectivity; other VLANs are fine. What are the two most likely causes, and which command separates them?

Either the VLAN is missing from a trunk's allowed list, or it has been deleted or shut down on one switch. `show interfaces trunk` separates them — read `Vlans allowed` and `active`, which reflects both the allowed list and whether the VLAN exists.

CHECKPOINT 2 of 3

A port is errdisabled. You issue no shutdown and it errdisables again within seconds. What have you forgotten?

Discuss · then advance

Checkpoint 2 of 3

A port is errdisabled. You issue no shutdown and it errdisables again within seconds. What have you forgotten?

To fix the cause. Err-disable is a symptom; whatever triggered it — a port security violation, a BPDU on a PortFast port — is still there and will trigger again the moment the port comes up.

CHECKPOINT 3 of 3

Why does knowing what *still works* narrow a fault faster than knowing what is broken?

Discuss · then advance

Checkpoint 3 of 3

Why does knowing what *still works* narrow a fault faster than knowing what is broken?

Because everything that still works is evidence you can trust, and it eliminates whole layers and paths at once. A user's report tells you one thing is broken; knowing that four similar things are fine tells you where the fault *cannot* be, which is a much larger reduction in the search space.

What to take away

- Establish scope first — one device, one VLAN, one switch, or everything. Scope points at the layer before you type anything.
- Then work bottom-up: port up, port clean, right VLAN, MAC learned, VLAN on the trunk, spanning tree forwarding, gateway present.
- Read `show logging` early. It is timestamped and often explicit.
- Six faults cover most cases: wrong VLAN, VLAN missing from a trunk, native VLAN mismatch, errdisable, spanning tree blocking, duplex mismatch.
- Capture a baseline while the network works. Change one thing at a time, and remove the cause before recovering an errdisabled port.

Where we got to

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NEXT

Chapter 15 — The Routing Table and How a Router Decides